TECHNICAL OVERVIEW

SignFlow

Real-time sign language captioning that runs entirely on an ordinary laptop. Point it at a webcam or any window on screen, and it turns signing into written English — without sending a single frame anywhere.

2,731	81.35%	0.17 s	0
SIGNS RECOGNISED	TOP-1 ACCURACY	PER SIGN	DATA LEAVES DEVICE

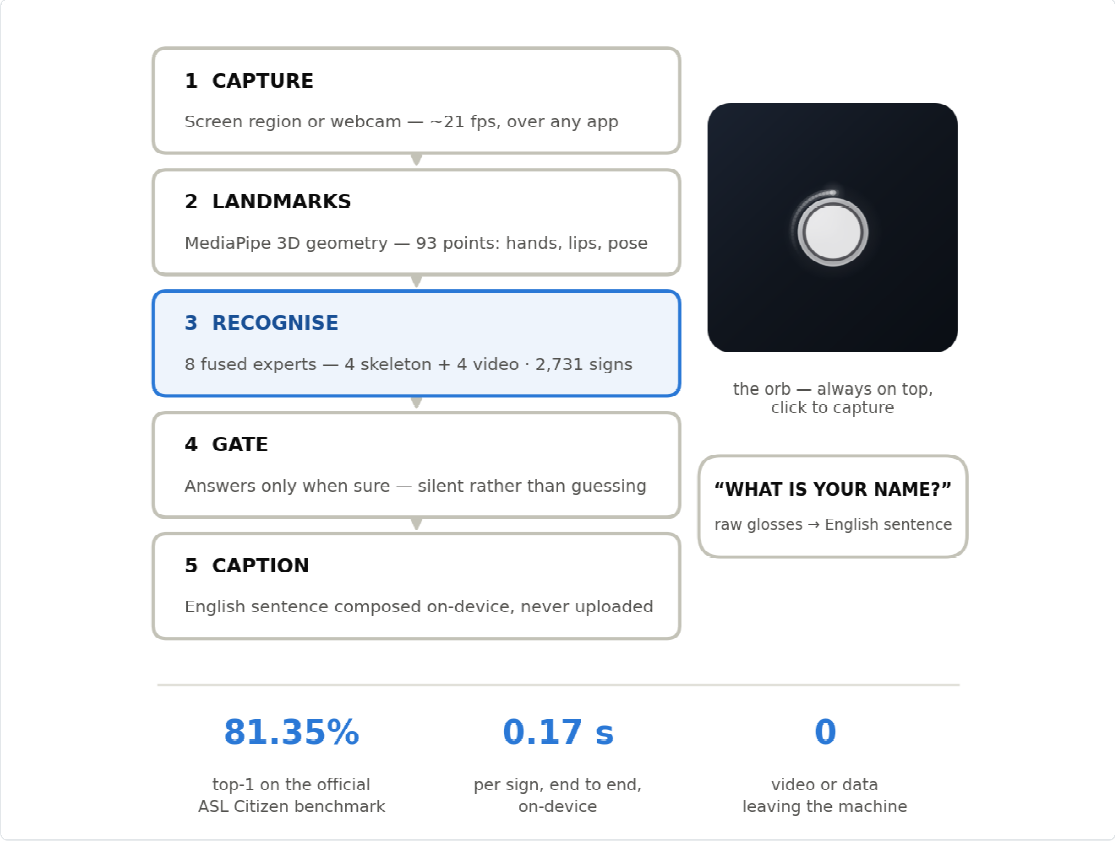
What problem it solves

Around 70 million people use a sign language as their first language. Almost nobody else understands one. That leaves a Deaf person dependent on a human interpreter — someone who must be booked, paid, and physically present — for a doctor's appointment, a job interview, or a conversation at a counter.

SignFlow removes the interpreter from the simple cases. A hearing person opens it, the Deaf person signs, and English appears. It works over a video call, over a recorded lecture, or face to face through the webcam.

Because it runs on-device, it can be used where a cloud service cannot: a clinic discussing a diagnosis, a bank discussing an account, a classroom with no reliable internet. No video is uploaded, so there is nothing to leak.

How it works



The five stages, from pixels to an English sentence

1 **Capture.** A screen region or the webcam is sampled at roughly 21 frames per second. The orb floats above every other window, so it works over any application — a video call, a media player, a browser.

2 **Landmarks.** MediaPipe extracts 93 three-dimensional points per frame: 21 per hand, 40 around the lips, and 10 across the upper body. Sign meaning lives in all three — handshape, mouth pattern, and where the hands sit relative to the body.

3 **Recognition.** Eight independent models score the clip and their opinions are combined. This is the core of the system and is described below.

4 **Gate.** The answer is released only when the winner is clearly ahead of the runner-up. On an unclear clip SignFlow stays silent rather than guessing — roughly one clip in eleven is declined.

5 **Caption.** Recognised signs are ordered into an English sentence on-device. Sign languages have their own grammar, so this is a translation step, not a word swap.

Why eight models instead of one

SignFlow reads the same clip two fundamentally different ways, then combines them.

The skeleton stream — geometry

A transformer reads the sequence of 93 landmark positions over time. It sees pure motion and shape: where the fingers are, how the hands travel, what the mouth does. Because it never sees pixels, it is unaffected by lighting, clothing, skin tone, or background.

The appearance stream — pixels

A pretrained video transformer reads the raw image. It catches what landmarks lose: fine handshape detail, fingers touching, fast motion that blurs, and moments when tracking drops a hand entirely.

Why combining them works

The two families make *different mistakes*. Measured across 32,940 clips, a skeleton model and an appearance model disagree in their errors far more than two models of the same kind do. That disagreement is exactly what makes combining them pay:

CONFIGURATION	ACCURACY
Best single skeleton model	**71.38%**
Skeleton + one appearance model	**79.45%**
All eight models combined	**81.35%**

Adding one *different* model gains eight points. Adding six more of the same kinds gains under two. The system is designed around that fact.

> **Trained to survive the real world.** An early version scored 72% on its own benchmark and 2% on video from a different source — it had learned one dataset's cameras rather than signing itself. Retraining across four separate recording setups lifted real-world accuracy to 65% and improved benchmark accuracy at the same time. Every model is now measured both ways before it ships.

The interface

SignFlow is a single floating orb. It sits above everything, takes almost no screen space, and has no window chrome, menu bar, or dock icon.

The orb — click to capture

Three gestures

Click	Capture and caption
Long press	Draw a screen region
Double tap	Open the tool wheel

The ring around the orb reports state: it pulses while capturing and traces a comet while the model is thinking, so the user always knows whether it is listening.

Features

Double-tap the orb to open the wheel — every action in one gesture

FEATURE	WHAT IT DOES
Screen	Captions signing in any window — a video call, a recorded lecture, a social media clip. Works over applications that have no accessibility support of their own.
Region	Restricts capture to a drawn rectangle, so only the signer is read and the rest of the screen is ignored.
Webcam	Face-to-face mode for in-person conversation.
Captions	Controls the caption pill — position, size, and how long text stays on screen. The pill remembers its corner between sessions and is clamped back on screen if a display is unplugged.
Theme	Light and dark presentation for the overlay.
Settings	Capture rate, confidence threshold, and model selection.
Close Menu	Collapses the wheel back to the orb.
Exit Orb	Quits cleanly and releases the camera.

Beyond the wheel

Continuous mode	Rather than captioning one sign at a time, SignFlow watches a stream of signing, detects where each sign begins and ends from the motion itself, and captions each one as it completes.
Silence when unsure	The system releases an answer only when the leading candidate is clearly ahead of the second. For an accessibility tool, a confident wrong answer is worse than no answer.
Sentence composition	Recognised signs are assembled into English with a guard that refuses to invent content the signer did not produce — the model may reorder and add grammar, but not add meaning.
Signing avatar	The reverse direction: typed text is performed by a rigged 3D character, so a hearing person can reply in sign. Currently a working prototype.
Fully offline	Every model runs locally. No account, no network, no per-minute cost — and nothing to intercept.

Where it stands

MEASURE	VALUE
Signs recognised	**2,731**
Accuracy, top-1	**81.35%**
Accuracy, correct answer in top 5	**96.30%**
Evaluated on	32,940 clips from signers excluded from training
Speed	**0.17 s per sign**
Hardware required	One consumer laptop GPU

Accuracy is measured on people the system has never seen. Testing on familiar signers inflates results substantially, so every figure here is the harder number.
